

Linear Algebra

In-Depth Study Notes

Chapter 2, Part 1 — Linear Systems, Gaussian Elimination & Echelon Form

How to Read These Notes, and Why “Part 1”

Chapter 2 in your source is roughly twice the length of Chapter 1, covering everything from “what is a linear system” through determinants and Cramer’s rule. Cramming all of it into one document would mean rushing the computational skills that everything later in the course depends on –so, exactly as before, it’s split. **Part 1** (this file) covers §2.1–2.4: what a linear system is, how to solve one by hand (Gaussian elimination), and how to package that into the row reduced echelon form and elementary matrices. **Part 2** (next time you ask) will cover §2.5–2.8: rank, the existence/uniqueness theorem, invertible matrices, and determinants.

Colour code, same as before:

- **Blue** = original *Definition*, restated precisely.
- **Purple** = original *Theorem / Result*.
- **Green** = “In Plain English.”
- **Grey** = a fully worked example (fresh numbers throughout).
- **Orange** = a practice problem for you to try.
- **Coloured bold text** = a jargon word, explained on first use.

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1 §2.1–2.2 What Is a Linear System?

The geometric picture

Before any formulas: a single linear equation $ax + by = c$ (with a, b not both 0) draws a **line** in the plane. Two such equations describe two lines, and solving the system means finding where they meet. There are exactly three possibilities:

In Plain English – three outcomes, geometrically

- **Unique solution:** the two lines cross at exactly one point (e.g. $x + 2y = 1$ and $x + 3y = 1$ meet only at $(1, 0)$).
- **Infinitely many solutions:** the two equations secretly describe the *same* line (e.g. $x + 2y = 1$ and $2x + 4y = 2$ – the second is just twice the first), so every point on that line is a solution.
- **No solution:** the lines are parallel and distinct (e.g. $x + 2y = 1$ and $2x + 4y = 3$ – same direction, different line, so they never meet).

The same three outcomes happen with three equations in three unknowns, now read as three **planes** in space (a unique solution = all three planes meet at one point; infinitely many = they meet along a whole line; no solution = no common point at all, e.g. three mutually “parallel-ish” planes with no shared intersection). This 3-way split is not a coincidence of small examples – §2.6 will prove it is the *only* possible outcome for *any* linear system, of any size.

Formal definitions**Definition 2.2.1 – Linear System**

A **linear system** of m equations in n unknowns $x_1, \dots, x_n$ is a set of equations

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2, \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m, \end{aligned}$$

with $a_{ij}, b_i \in \mathbb{R}$. Writing $A = [a_{ij}]$ (the **coefficient matrix**), $x = (x_1, \dots, x_n)^t$, and $b = (b_1, \dots, b_m)^t$, this is compactly $Ax = b$. The block matrix $[A \ b]$ is the **augmented matrix** of the system. The system is **homogeneous** if $b = 0$ (every $b_i = 0$), and **non-homogeneous** otherwise.

Definition 2.2.3 – Solution

A **solution** of $Ax = b$ is a column vector $y = (y_1, \dots, y_n)^t$ such that substituting $x_i = y_i$ satisfies every equation, i.e. $Ay = b$. The system $Ax = 0$ always has $x = 0$ as a solution – called the **trivial solution**. Any solution $x \neq 0$ of $Ax = 0$ is called a **non-trivial solution**.

In Plain English

The coefficient matrix A is just “the numbers multiplying the unknowns,” stripped of the unknowns themselves; the augmented matrix glues the right-hand side on as one extra column, so the *entire* system – every number that matters – lives in one rectangular table. This is deliberate: once we’re solving systems, we will manipulate the augmented matrix directly and never need to rewrite $x, y, z, \dots$ or the “=” signs at every step.

A Preview of the Solution Method

Consider $x + 7y + 3z = 11$, $x + y + z = 3$, $4x + 10y - z = 13$. The idea (formalised as **Gaussian elimination** in §2.3) is to use one equation to eliminate a variable from the others, repeatedly, until only one unknown remains in the last equation, then work backwards. This “eliminate, then back-substitute” strategy is the entire content of the next two sections – just made completely mechanical by working with the augmented matrix instead of full equations.

Practice Problems – §2.1–2.2

- 2.1.1 Write $2x - 3y + z = 5$, $x + y - 4z = 0$ in the form $Ax = b$, clearly identifying A , x , and b .
- 2.1.2 Is the system in Problem 1 homogeneous? If not, write down its **associated homogeneous system** (same A , but b replaced by 0).
- 2.1.3 Without solving it, is $x = (1, 1, 1, 1)^t$ a solution of $x_1 + x_2 - x_3 - x_4 = 0$, $2x_1 - x_2 + x_3 + 2x_4 = 6$? (Just substitute and check both equations.)
- 2.1.4 Sketch (mentally or on paper) two lines that meet at exactly one point, then perturb one line slightly so that they become parallel. What happened to the number of solutions?
- 2.1.5 (*Challenge*) Can a system of *one* linear equation in *two* unknowns ever have a unique solution? Explain geometrically.

2 §2.3 Row Operations, Equivalent Systems, and Gaussian Elimination

Why elimination preserves solutions

Definition 2.3.1 – Elementary Operations (on equations)

- (1) Interchange two equations.
- (2) Multiply an equation by a nonzero constant c .
- (3) Replace an equation by itself plus c times another equation.

Definition 2.3.2 – Equivalent Linear Systems

Two linear systems are **equivalent** if one can be obtained from the other by a finite sequence of elementary operations.

Lemma 2.3.3 & Theorem 2.3.4

A *single* elementary operation never changes the solution set of a system (Lemma 2.3.3). Applying elementary operations *repeatedly* therefore also never changes the solution set (Theorem 2.3.4, proved by induction on the number of operations applied): **equivalent systems have exactly the same solutions**.

In Plain English – why is this true?

Every elementary operation is **reversible**: swapping two equations back undoes a swap; dividing by c undoes multiplying by c ; subtracting c times an equation undoes adding c times it. Because each step can be perfectly undone, no solution can ever be lost, and (since real-number arithmetic doesn't invent new equalities out of nowhere) no new, spurious solution can be introduced either. This reversibility is *the* reason we're allowed to transform a hard system into an easy one and trust that solving the easy one solves the original.

Definition 2.3.5 – Elementary Row Operations

The matrix version of Definition 2.3.1, applied to rows of the augmented matrix instead of whole equations:

- R_{ij} : interchange row i and row j .
- $R_k(c)$: multiply row k by a nonzero constant c .

- $R_{kj}(c)$: replace row k by $(\text{row } k) + c \cdot (\text{row } j)$.

Two matrices are **row equivalent** if one is obtained from the other by a finite sequence of elementary row operations.

Gaussian elimination in action

Definition 2.3.9 – Gaussian (Forward) Elimination

Gaussian elimination solves $Ax = b$ by applying elementary row operations to the augmented matrix $[A \ b]$ until it reaches **upper triangular form** (all entries below the diagonal are 0), then solving from the bottom equation upward – called **back substitution**.

Worked Example 1 – a unique solution

Solve $x + y + z = 6$, $2x + y - z = 1$, $x - y + 2z = 5$.

$$[A \ b] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 1 & -1 & 1 \\ 1 & -1 & 2 & 5 \end{array} \right].$$

Step 1 ($R_{21}(-2)$, $R_{31}(-1)$: eliminate x from rows 2,3 using row 1):

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -1 & -3 & -11 \\ 0 & -2 & 1 & -1 \end{array} \right].$$

Step 2 ($R_{32}(-2)$: eliminate y from row 3 using row 2):

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -1 & -3 & -11 \\ 0 & 0 & 7 & 21 \end{array} \right].$$

This is upper triangular. **Back-substitute**: row 3 gives $7z = 21 \Rightarrow z = 3$. Row 2 gives $-y - 3(3) = -11 \Rightarrow y = 2$. Row 1 gives $x + 2 + 3 = 6 \Rightarrow x = 1$. **Unique solution**: $(x, y, z) = (1, 2, 3)$.

Worked Example 2 – infinitely many solutions

Solve $x + y + z = 4$, $2x + 2y + 2z = 8$, $x - y + z = 2$. Eliminating using row 1 ($R_{21}(-2)$, $R_{31}(-1)$):

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & -2 & 0 & -2 \end{array} \right].$$

Row 2 became **entirely zero** – it carried no new information (it was exactly twice row 1 to begin with). Swapping rows 2 and 3 for convenience and reading off: row “ $0, -2, 0 \mid -2$ ” gives $y = 1$; row 1 gives $x + 1 + z = 4 \Rightarrow x = 3 - z$. With z left completely free, the solution set is

$$\{(x, y, z) = (3 - z, 1, z) : z \in \mathbb{R}\},$$

an **infinite family** of solutions, one for every choice of z .

Worked Example 3 – no solution

Solve $x + y + z = 4$, $2x + 2y + 2z = 9$, $x - y + z = 2$ (identical to Example 2, except the right-hand side of the second equation is now 9 instead of 8). Eliminating exactly as before:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 0 & 0 & 1 \\ 0 & -2 & 0 & -2 \end{array} \right].$$

The second row now reads $0x + 0y + 0z = 1$ – **impossible** for any x, y, z . As soon as a row of the form $[0 \ 0 \ \cdots \ 0 \ | \ c]$ appears with $c \neq 0$, stop: the system has **no solution**. (Geometrically: the same three planes as Example 2, except now the second plane has been shifted just enough that all three no longer share a common point.)

The Pattern to Remember

Once a system is in upper triangular (or fully reduced) form, exactly one of three things happens, matching the geometric picture from §2.1:

- every variable gets pinned down by back-substitution $\Rightarrow$ **unique solution**;
- a row becomes entirely zero (including the right-hand side) $\Rightarrow$ a variable stays free $\Rightarrow$ **infinitely many solutions**;
- a row becomes $[0 \ \cdots \ 0 \ | \ c]$ with $c \neq 0 \Rightarrow$ **no solution**, no matter what happens elsewhere.

Practice Problems – §2.3

- 2.3.1** Solve $x + y = 5$, $2x - y = 1$ by Gaussian elimination, showing every row operation you use.
- 2.3.2** Solve $x + 2y + z = 3$, $2x + 4y + 2z = 6$, $x - y + z = 0$. (Watch for a zero row.)
- 2.3.3** Solve $x + y = 2$, $x + y = 5$. What goes wrong, and how does it show up in row-reduced form?
- 2.3.4** Name the single elementary row operation that undoes $R_{23}(4)$ (replace row 2 by row 2 + 4 × row 3).
- 2.3.5** (*Challenge*) Explain why $R_k(0)$ (multiplying a row by 0) is *deliberately excluded* from the list of elementary row operations in Definition 2.3.5, using Lemma 2.3.3's reversibility argument.

3 §2.4 Row Reduced Echelon Form and Elementary Matrices

Gaussian elimination (stopping at upper-triangular form) is enough to solve a system, but pushing the row reduction *all the way* gives an even cleaner, completely standardised form – one where the answer can simply be read off.

Row reduced form and echelon form**Definition 2.4.1 & 2.4.3 – Row Reduced Form, Leading Terms**

A matrix C is in **row reduced form** if (1) the first nonzero entry in each row is 1 (called a **leading term**), and (2) the column containing that leading 1 (the **leading column**) has *every other* entry equal to 0.

Definition 2.4.4 – Basic and Free Variables

Once $[A \ b]$ is row-reduced to $[C \ d]$, the variables corresponding to leading columns (among the first n columns) are **basic variables**; all other variables are **free variables** – they may be assigned any value, with the basic variables then determined in terms of them.

Definition 2.4.6 – Row Reduced Echelon Form (RREF)

A matrix is in **row reduced echelon form** if, in addition to being row reduced, (1) any all-zero rows sit at the very bottom, and (2) the leading terms move strictly left to right as you go down the rows.

In Plain English

RREF is the “neatest possible” version of a matrix under row operations: every pivot (leading entry) is a clean 1, isolated in its own column, with zero rows pushed to the bottom and the staircase of pivots marching rightward as you read down. This is the standardised endpoint every augmented matrix can be driven to – and remarkably (Theorem 2.4.11, stated without proof in the source), **it’s unique**: no matter which sequence of row operations you choose, you always land on the exact same RREF.

Definition 2.4.9 – Gauss–Jordan Elimination

Gauss–Jordan elimination pushes a matrix all the way to RREF: first forward (Gaussian) elimination down to upper-triangular form, then **back substitution** carried out as *row operations* (clearing entries *above* each pivot too, not just below).

Worked Example – continuing to RREF

Continuing Worked Example 1 above, we had reached $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -1 & -3 & -11 \\ 0 & 0 & 7 & 21 \end{array} \right]$. First, scale each row so every leading entry is 1 ($R_2(-1)$, $R_3(1/7)$):

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 3 & 11 \\ 0 & 0 & 1 & 3 \end{array} \right].$$

Now clear *above* each pivot. Using row 3 to clear column 3 in rows 1,2 ($R_{13}(-1)$, $R_{23}(-3)$):

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right].$$

Using row 2 to clear column 2 in row 1 ($R_{12}(-1)$):

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right].$$

This is RREF – and the solution $x = 1$, $y = 2$, $z = 3$ (matching Worked Example 1) is now sitting in the last column, with no back-substitution arithmetic required at all. This is the real payoff of pushing all the way to RREF.

Elementary matrices

Definition 2.4.13 – Elementary Matrix

An **elementary matrix** is the matrix obtained by applying exactly *one* elementary row operation to the identity matrix I_n . Matching Definition 2.3.5, there are three types: E_{ij} , $E_k(c)$, $E_{ij}(c)$.

The Key Fact About Elementary Matrices

For any matrix A and elementary row operation e , if $E = e(I)$ is the corresponding elementary matrix, then $e(A) = EA$: **applying a row operation to A is the same as left-multiplying A by the matching elementary matrix.** Repeating this, Gaussian/Gauss–Jordan elimination on A is really just left-multiplying A by a whole sequence of elementary matrices, one per row operation performed.

In Plain English

This is a genuinely important shift in viewpoint: row-reducing a matrix by hand *feels* like a sequence of scribbled arithmetic steps, but Theorem above says it is secretly just repeated matrix multiplication, $E_k \cdots E_2 E_1 A$. This is the bridge that connects the hands-on computational technique of §2.3–2.4 to the abstract algebra of invertible matrices in §2.7 (next time): once we know elimination is multiplication by special matrices, and those special matrices turn out to always be invertible, a great deal falls into place almost for free.

Worked Example – elementary matrices in action

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$.

- $E_{23} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ (swap rows 2,3 of I_3). Then $E_{23}A = \begin{pmatrix} 1 & 2 & 3 \\ 7 & 8 & 9 \\ 4 & 5 & 6 \end{pmatrix}$ – rows 2 and 3 of A have swapped, exactly as R_{23} would do by hand.
- $E_2(3) = \text{diag}(1, 3, 1)$ (scale row 2 of I_3 by 3). Then $E_2(3)A = \begin{pmatrix} 1 & 2 & 3 \\ 12 & 15 & 18 \\ 7 & 8 & 9 \end{pmatrix}$ – row 2 of A has been tripled.
- $E_{13}(2) = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ (row 1 of I_3 plus $2 \times$ row 3). Then $E_{13}(2)A = \begin{pmatrix} 15 & 18 & 21 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$ – row 1 has become (row 1) + $2 \times$ (row 3) = $(1 + 14, 2 + 16, 3 + 18) = (15, 18, 21)$, while rows 2 and 3 are untouched.

Practice Problems – §2.4

- 2.4.1** Row-reduce $\begin{pmatrix} 2 & 4 & 2 \\ 1 & 1 & 0 \end{pmatrix}$ all the way to RREF, and identify the basic and free variables if this were the coefficient matrix of a homogeneous system.
- 2.4.2** Write down the RREF of $\begin{pmatrix} 0 & 1 & 3 \\ 0 & 0 & 0 \\ 1 & 2 & 5 \end{pmatrix}$. (*Hint:* start by getting a nonzero entry into the top-left position.)

- 2.4.3** Write out the elementary matrix $E_3(-2)$ of order 3 (scale row 3 by -2), then compute $E_3(-2)B$ for $B = \begin{pmatrix} 1 & 0 \\ 2 & 1 \\ 3 & 4 \end{pmatrix}$, and check the result matches what the row operation should do by hand.
- 2.4.4** Write out $E_{12}(5)$ of order 2, and verify by direct computation that $E_{12}(-5)$ is its inverse, i.e. $E_{12}(-5)E_{12}(5) = I_2$.
- 2.4.5** (*Challenge*) A matrix has RREF equal to I_3 . What does this tell you about the number of solutions of $Ax = b$ for *any* choice of $b \in \mathbb{R}^3$? (This is a first glimpse of §2.7's invertibility theory.)

Part 1 Wrap-Up

What You Now Know

- How to package a linear system as $Ax = b$, and what the coefficient and augmented matrices are.
- Why elementary row operations never change the solution set (reversibility), which is what makes row-reduction trustworthy at all.
- Gaussian elimination by hand, and how to read off unique / infinite / no-solution outcomes directly from the shape of the reduced rows.
- Row reduced echelon form (RREF) – the unique “neatest” endpoint of row reduction – and Gauss–Jordan elimination to reach it.
- Elementary matrices, and the key fact that every row operation is secretly a left-multiplication by one of them.

Self-Check Before Part 2

You should be able to, without looking anything up:

1. Row-reduce a 3×3 augmented matrix to RREF by hand and state the solution set (unique / infinite / none) directly from the result.
2. Explain, in one sentence, why elementary row operations don't change the solution set.
3. Write down the elementary matrix for a given row operation, for all three types.

If any of these feel shaky, revisit that box above – Part 2 (rank, existence of solutions, invertibility, determinants) builds directly on all three, starting from its very first page.

That's Part 1. Message me whenever you're ready and I'll build Chapter 2, Part 2 – §2.5–2.8: Rank, Existence & Uniqueness of Solutions, Invertible Matrices, and Determinants – in the same format.